Horizontal launches

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Find flight time from vertical motion and then horizontal range.

SUCCESS CRITERIA - I can...

1. Separate vertical and horizontal givens.
2. Find or use fall time before calculating range.
3. Check units and interpret the landing point.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “horizontal” and “vertical” to explain your answer. Sentence starters are optional.

horizontal: along a level line, usually the x direction

vertical: up and down, usually the y direction

displacement: change from initial position to final position, with direction

01 Warm-up: bring back an earlier idea

A spherical planet has surface field **75 N/kg**. **Find the field** at center distance **5** R, where R is the planet radius.

Word help: radius = distance from the center to the edge of a circle or sphere

Sentence starter: My design includes … . It does what is needed by … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $h = \frac{1}{2}gt^2$; range = $v_x t$.
Horizontal launch; h is the positive drop, t is fall time; g = **10 m/s²**. CPA may use the supplied time.

Worked reasoning: A horizontal launch falls **20 m** with g=**10 m/s²**. From **20**=**5** t², t=**2 s**. At horizontal speed **3 m/s**, range=**6 m**.

Your turn: Draw a diagram labeling height and range. Number the two solution steps to **show why time comes first**.

Sentence starter: I think … . I can point to …, which shows …, so … .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

On the diagram, label height h, horizontal range and initial horizontal velocity. **Which direction** determines the fall time?

Sentence starter: I found … [unit]. This value means … in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

04 Apply the physics

In a launch simulation, two balls leave the same platform horizontally. Both fall for **2 s**. One launches at **4 m/s** and the other at **6 m/s**. Draw and **label both paths**; **calculate each horizontal range**.

Sentence starter: I found … [unit]. This value means … in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Two platforms have heights **5** and **20 m**. Supplied fall times are **1** and **2 s**. At the same launch speed, does quadrupling height quadruple range? Explain.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

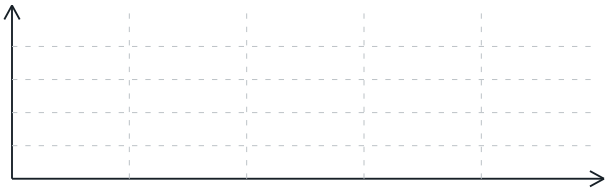
06 Review: connect an earlier idea

A **6 kg** cart starts from rest under **48 N** net force right for **5 s**. **Find acceleration** and final velocity. **State one thing you** are treating as true.
Label the given and predicted quantities on the diagram.

Word help: net force = the vector sum of all forces on one object

Sentence starter: I predict … . I used … to make this prediction because … .

Velocity (m/s)



Time (s)

Marzano check - circle: 1 2 3

07 Exit check

A horizontal launch at **5 m/s** stays in air **3 s**. Where does it land relative to the platform base? **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.

Sentence starter: In my work, I used … to show … . I labeled it … .

Separate horizontal and vertical motion

Label x and y; state positive directions.

Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A target is **10 m** from a platform whose fall time is **2 s**. **Choose the horizontal launch speed**, then **predict where half** that speed lands. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.

Sentence starter: I predict … . I used … to make this prediction because … .



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3